The research you are not doing

CMU Africa Research Club Talk

Steven Kolawole

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PoA

- Opening: What you know through the flier
- Background – The Unlikely Origin Story
- First Project: The Sign Language Fumble That Worked
- Current Research: Making AI Less Elitist
- The PhD Reality Check
- Why Research matters, **even if You Don’t Want a PhD**
- The CMU Advantage You’re Not Using
- Practical Research Starter Pack
- Leveraging CMU Africa’s Strategic Position
- The Meta-Research Game
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Opening



Opening

- *PhD student,* Carnegie Mellon University (2023 -), Language Technologies Institute, School of Computer Science, CMU
  - Works with Prof. Virginia Smith, Machine Learning Department, SCS, CMU
- *BSc, Computer Science,* Federal University of Agriculture Abeokuta, Nigeria (2017-2023)
  - *Got in straight from undergrad*
    - *Had an IJCAI paper + 2 small workshop papers*

The Numbers

- *700+ PhD students in CMU's School of Computer Science*
- *~1-1.5% are Black (that's about 7-10 people total)*
- *I started at FUNAAB (Federal University of Agriculture Abeokuta) - not exactly a typical CMU pipeline*
  - *Didn’t even finish with a first class*
  - *Had a carryover course in Linear Algebra II*

Other background

Quick backstory (because origin stories matter)

- *Discovered computers at 18 (I'm not your typical "coded since age 5" story)*
- *Worked in a block industry from age 10 to pay for school*
- *FUNAAB undergrad wondering how to get to the global ML scene*

- ***The Real Lesson:*** *You don't need perfect conditions to start. Exposure matters a lot, and CMU Africa offers that*

The Sign Language Fumble That Worked

- **The Problem:** Zero datasets for sub-Saharan African sign languages. Literally zero.
- **The "Solution"**: Built the first Nigerian sign language dataset with teammates across 4 countries

- Me in Abeokuta, teammate in Lagos, collaborator in Toronto, mentor in Cardiff
- Leveraged on open science communities (Masakhane, ML Collective)
- Completely remote, completely scrappy
- Brought along a list of good fortune (first paper, lots of awards, etc)

The Sign Language Fumble That Worked

Sign-to-Speech Model for Sign Language Understanding: A Case Study of Nigerian Sign Language

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] 2 Nov 2021

Current Research: Making AI Less Elitist

My central question: *"How can intelligent algorithmic design eliminate unnecessary resource requirements in AI systems?"*

Research 1: Agreement-Based Cascading (ABC)

Published in Transactions on Machine Learning Research (07/2025)

Agreement-Based Cascading for Efficient Inference

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Reviewed on OpenReview: *https://openreview.net/forum?id=jn9B7LMlzk*

Research 1: Agreement-Based Cascading (ABC)

- Problem: Why use GPT-4 when GPT-3.5 can handle the task?
- Solution: When small models agree, trust them. When they disagree, defer to larger models.
- Result: 2-25× cost reduction while often improving accuracy
- *Translation: Smart routing beats brute force*

Research 1: Agreement-Based Cascading (ABC)

Training-Free Semantic Deferrals for Open-Ended LLM Cascades

Duncan Soiffer [1] **Steven Kolawole** [1] **Virginia Smith** [1]

Abstract

Existing cascade systems struggle with open-ended text generation due to evaluation challenges where multiple valid outputs exist without ground truth references. We propose using semantic agreement between multiple model outputs as a training-free deferral signal and evaluate semantic similarity metrics against token-level confidence across translation, summarization, question answering, and reading comprehension

model outputs as a training-free deferral signal. When several smaller models produce semantically consistent answers, their collective response likely represents reliable output. Conversely, semantic disagreement suggests uncertainty, indicating deferral would be beneficial. By focusing on output agreement rather than learned confidence, our approach eliminates supervision while remaining applicable across diverse tasks.

We systematically explore semantic agreement capturing

Research 2: Forward-Pass Pruning

Preprint. Under review.

Everybody Prune Now: Structured Pruning of LLMs with Only Forward Passes

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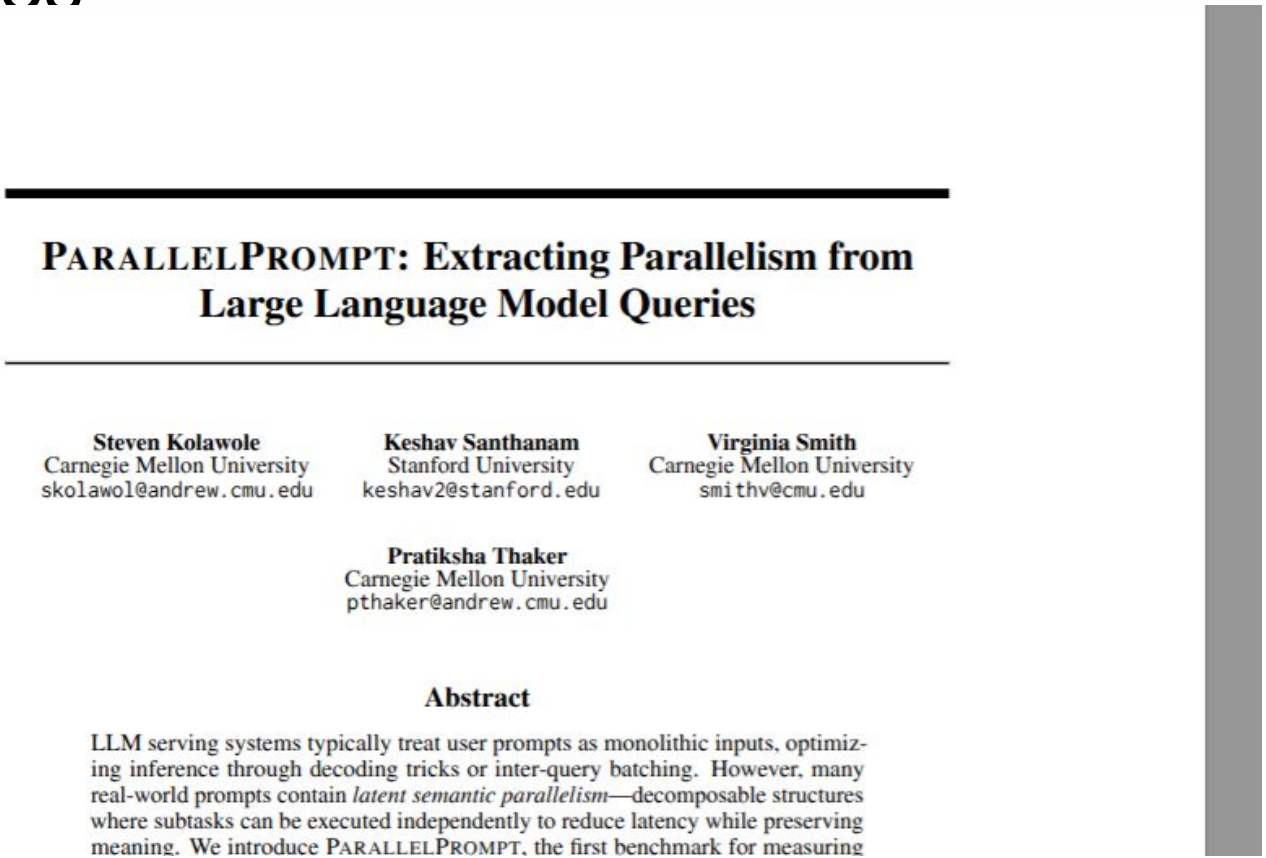
Abstract

Structured pruning is a promising approach to create smaller, faster LLMs. However, existing methods typically rely on backward passes, which can

Research 2: Forward-Pass Pruning

- Problem: Pruning LLMs requires 40-80GB GPUs (basically, you need to be Google)
- Solution: Prune models using only forward passes
- Result: Compress LlaMA-7B on a 24GB GPU (3× less memory than gradient methods)
- *Translation: Model optimization for the rest of us*

Research 3: Extracting Parallelism in LLM Queries



PARALLELPROMPT: Extracting Parallelism from Large Language Model Queries

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Abstract

LLM serving systems typically treat user prompts as monolithic inputs, optimizing inference through decoding tricks or inter-query batching. However, many real-world prompts contain *latent semantic parallelism*—decomposable structures where subtasks can be executed independently to reduce latency while preserving meaning. We introduce PARALLELPROMPT, the first benchmark for measuring

[cs.LG] 26 Jun 2025

Research 3: Extracting Parallelism in LLM Queries

- Problem: We execute queries sequentially even when they could run in parallel
- Discovery: 10% of natural user queries contain latent parallelism
- Result: Up to 5× speedups through semantic decomposition
- *Translation: Make LLMs faster without touching hardware*

Current Research: Making AI Less Elitist

The Meta-Point: Those "necessary" barriers (memory requirements, training overhead, sequential execution) are often just poor design choices.

Core Thesis: *"Scarcity in AI is not inevitable."* Performance doesn't have to come at the cost of accessibility.

The PhD Reality Check

The Honest Truth

- High workload (research + coursework + teaching + imposter syndrome)
- High Pressure (CMU is a top 3 school for CS; highly competitive)
- Being part of that <1.5% while representing for the culture
- Constantly questioning if you belong in rooms where you're the only one

The PhD Reality Check

What Actually Helped:

- Communities: I had 2 black guys + 1 Indian guy that helped
- Proactive networking: Don't wait for invitations
- Asking for help: Turns out, people actually want to help if you ask properly
- **Locking in like crazy**

Lesson 1: Support systems matter as much as technical skill. Research is inherently collaborative.
Lesson 1: You will work your derriere off!

Why Research Matters for You

Industry Impact
Research teaches problem framing, not just solution implementation

- Industry roles increasingly value research thinking
- Product teams need people who can identify which problems are worth solving
- R&D roles are proliferating (look at OpenAI, Anthropic, Google DeepMind)
- AI *might* take away your entry level job

Why Research Matters for You

Publications = Credibility Currency

- Your GitHub shows you can code
- Your publications show you can think
- Recruiters increasingly filter for research experience

Why Research Matters for You

Global Collaboration Skills & Network Effects

- Research communities transcend geography
- Remote async collaboration is the future of work
- You're already practicing this at CMU Africa
- One good paper = speaking invites = job opportunities = more collaborations
- *Compound interest, but for careers*

The CMU Advantage You’re Not Using

You have access to:

- CMU Pittsburgh faculty (virtually)
- CMU's industry partnerships
- Alumni network across all campuses
- Research computing resources
- Seminar series and reading groups

"Don't think local, think global. Your `@andrew.cmu.edu` works everywhere."

Practical Research Starter Pack

Find Your Niche (The Anti-Hype Strategy)

- Don't chase what's trending on Twitter/X
- Look for problems where your context is an advantage
- African datasets, deployment constraints, resource limitations = research gold

Example: *My efficiency research comes from experiencing resource constraints firsthand.*

Practical Research Starter Pack

Start Small, Think Big

- Reproduce a paper (seriously, most can't be reproduced)
- Improve one small thing (different dataset, better baseline, faster implementation)
- Submit to workshops (lower bar, faster feedback, same venue)
    - Workshops are practice rounds
- Scale to conferences (ICLR, NeurIPS, ICML)
- Collaboration beats solo projects

Practical Research Starter Pack

Do it in the Public

- GitHub: Clean, reproducible code
- Blog posts: Explain your thinking process
- Internal writeups: Track failed experiments (they matter!)

Pro tip: Your future self will thank you for good documentation. They serve as evidence of your credibility too.

Practical Research Starter Pack

Geographic Arbitrage

- Unique problem access (languages, infrastructure, deployment)
- oin CMU Pittsburgh research groups virtually
- Attend online seminars and workshops
- Collaborate with main campus students on projects
- Use CMU's industry partnerships for internship placements
- Regular check-ins: Don't disappear for weeks

The Long Game

Research compounds:

- Better problem identification → Better solutions → Better impact → Better opportunities

Closing Call-to-Action

The Research Imperative

Your coursework teaches you to solve known problems.
Research teaches you to find problems worth solving.

In a world where ChatGPT can code, your value isn't in implementation - it's in knowing what to build and why.

Closing Call-to-Action

Challenge for This Month
Pick ONE:

1. Join a reading group (ML Collective, local CMU group)
2. Read a paper a week
3. Start a mini-dataset (Nigerian languages, local problem)
4. Reach out to a potential collaborator (main campus student, faculty)
5. Reproduce one paper (pick something you use regularly)

Final Thought

"Don't wait for permission to start research. Your value isn't in the resources you have now - it's in the questions you choose to ask."

Remember: Research is how you go from consumer of AI to creator of AI. And at CMU Africa, you're perfectly positioned to create AI that works for everyone, not just Silicon Valley.

Q&A

Thank you!

"Also available for roasting your research ideas and potential collabs. I am usually late with responses."